

Real Estate Market Dashboard

KPI & DAX Measures Dictionary

This document outlines the core Key Performance Indicators (KPIs) utilized in the housing market dashboard. It provides the business definition, strategic value, and the underlying DAX logic for each metric.

Time Intelligence & Growth Metrics

YoY Sales Growth

Description: Calculates the percentage change in total sales revenue compared to the exact same period in the previous year.

Business Value: Essential for identifying market expansion or contraction, neutralizing seasonal effects to reveal true market momentum.

```
YoY_Sales_Growth =  
VAR CurrentYearSales = CALCULATE(SUM(Housing[purchase_price]),  
    YEAR(Housing[date]) = YEAR(MAX(Housing[date])))  
VAR PrevYearSales = CALCULATE(SUM(Housing[purchase_price]), YEAR(Housing[date])  
    = YEAR(MAX(Housing[date])) - 1)  
RETURN  
    DIVIDE(CurrentYearSales - PrevYearSales, PrevYearSales, BLANK())
```

Sales of Last 12 Months (TTM)

Description: Computes the total aggregate sales revenue for the trailing 12-month period rolling backward from the most recent transaction date.

Business Value: *Smooths out short-term volatility and seasonal spikes, providing a clearer picture of long-term revenue trends.*

```
SALES OF LAST 12 MONTHS =  
CALCULATE(  
    SUM(Housing[purchase_price]),  
    DATESINPERIOD(Housing[date], MAX(Housing[date]), -12, MONTH)  
)
```

Total YTD (Year-to-Date)

Description: Accumulates the total sales revenue from the beginning of the current calendar year up to the latest date in the dataset.

Business Value: *Allows stakeholders to track current annual performance against yearly targets.*

```
TotalYTD = TOTALYTD(SUM(Housing[purchase_price]), Housing[date])
```

Pricing & Valuation Metrics

Median Sales Change

Description: Calculates the year-over-year percentage change in the *median* purchase price of properties, rather than the average or sum.

Business Value: *Highly resistant to outliers (e.g., massive luxury estate sales). Provides a more accurate reflection of what a "typical" buyer experiences in the market.*

```
MEDIAN_SALES_CHANGE =  
VAR PrevYear = CALCULATE(MEDIAN(Housing[purchase_price]), YEAR(Housing[date]) =  
    YEAR(MAX(Housing[date])) - 1)  
VAR CurrYear = CALCULATE(MEDIAN(Housing[purchase_price]), YEAR(Housing[date]) =  
    YEAR(MAX(Housing[date])))  
RETURN  
    DIVIDE(CurrYear - PrevYear, PrevYear, BLANK())
```

Avg Price/SQM (Square Meter)

Description: The average final purchase price divided by the property's square meterage.

Business Value: *The gold standard for normalizing property values. Allows for direct comparison of value density between different regions (e.g., City Center vs. Rural) regardless of property size.*

```
Avg Price/sqm = AVERAGE(Housing[sqm_price])
```

Volume & Contextual Metrics

Sales By Region (Context Modifier)

Description: Calculates total sales but intentionally ignores all filters except the geographic 'region'.

Business Value: *Used primarily as a denominator to calculate "percentage of regional total" for specific property types, regardless of user slice selections.*

```
Sales By Region =  
CALCULATE(  
    SUM(Housing[purchase_price]),  
    ALLEXCEPT(Housing, Housing[region])  
)
```

Units Sold In Latest Year and Quarter

Description: Counts the exact number of distinct properties sold exclusively within the most recent year and most recent quarter available in the dataset.

Business Value: A pure velocity metric indicating current market liquidity and recent buyer activity levels.

```
Units Sold In Latest year and Quarter =  
COUNTX(  
    FILTER(  
        Housing,  
        AND(  
            YEAR(Housing[date]) = YEAR(MAX(Housing[date])),  
            QUARTER(Housing[date]) = QUARTER(MAX(Housing[date]))  
        )  
    ),  
    Housing[house_id]  
)
```

Property Age

Description: Dynamically calculates the age of the property at the time of the sale.

Business Value: Allows analysts to segment properties into "New Build" vs. "Historic" to analyze depreciation or historical premiums.

```
Age = YEAR(Housing[date]) - Housing[year_build]
```